Learning a Model and Controlling with It: GP-MPC, DeePC and Koopman MPC on Three Simulated Plants

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Abstract

This report describes a small benchmark of data-driven predictive controllers. Three plants with a known structure but wrong parameters (a mass-spring-damper, an inverted pendulum and a planar quadrotor with unknown drag and wind) are controlled by five controllers under the same horizon, weights, input limits and soft state constraints: MPC on the wrong prior model, MPC on the true model (an oracle), MPC on the prior model plus a Gaussian-process residual (GP-MPC), regularised data-enabled predictive control (DeePC) and linear MPC on a Koopman (EDMD) lifted model. All learned components use the same 1500 noisy samples per seed. Over 10 seeds, the three learned controllers remove most of the steady-state offset that the prior model leaves. GP-MPC has the fewest constraint violations, the Koopman controller is the fastest by an order of magnitude, and DeePC is the weakest on the quadrotor. Everything here is simulation; the claims are limited to what the experiments measured.

1 Problem setup

Each plant is a continuous-time model $\dot{x} = f(x, u; \theta)$ simulated with RK4 (10 sub-steps per sampling period $T_s = 0.05\,\mathrm{s}$) under a zero-order-hold input. The prior parameters θ_0 differ from the true parameters $\theta^\star$ (Table 1). The state is measured with additive Gaussian noise. Every controller solves, at each step, a finite-horizon problem with horizon $N = 20$,

$$\min \sum_{k=0}^{N-1} \|x_k - r_k\|_Q^2 + \|u_k - u^{\mathrm{ref}}\|_R^2 + \|x_N - r_N\|_{Q_f}^2 + \rho_1 \mathbf{1}^\top s + \rho_2 \|s\|^2, \tag{1}$$

with hard input bounds and the state constraints $x^{\min} \le C x_k \le x^{\max}$ softened by slacks $s \ge 0$ ($\rho_1 = 10^3$, $\rho_2 = 10^4$). The input reference u^{ref} is the equilibrium input of the current set-point under the prior model (under the true model for the oracle). Constraint violations are measured afterwards on the true, noise-free state.

Table 1: Plants and model mismatch.

Plant	State / input	Mismatch (true vs prior)
Mass-spring-damper	(p, v) / F	k: 2.0 vs 0.8, c: 0.4 vs 0.1, m: 1.0 vs 0.7
Inverted pendulum	(θ, ω) / τ	m: 1.0 vs 0.6; viscous and Coulomb friction unknown
Planar quadrotor	$(p_x, p_z, \phi, v_x, v_z, \omega)$ / (T_1, T_2)	m: 1.0 vs 0.8; drag 0.4 and wind 0.6 N unknown

Data. The pendulum and the quadrotor are open-loop unstable, so the data are collected in closed loop: an LQR designed on the prior model tracks random set-points (changed every 25 steps) while a Gaussian dither is added to the input. Each dataset has 15 episodes of 100 steps (1500 samples). Seed s draws the dataset; the closed-loop noise uses seed $500 + s$.

2 Controllers

Nominal and oracle MPC. Multiple-shooting NMPC on the RK4 discretisation (one sub-step) of the prior or the true model, solved with IPOPT [10] through CasADi [1], warm started with the shifted previous solution.

GP-MPC. The mismatch enters through the accelerations, so one GP per velocity state learns the one-step error $\Delta v = v_{k+1} - F_0(x_k, u_k)_v$ from features $z = (x_k^{\mathrm{f}}, u_k)$, where x^{f} leaves out translation-invariant positions. The prediction model is $x_{k+1} = F_0(x_k, u_k) + E\,\mu(z_k)$ with E mapping Δv to the velocities and $\frac{T_s}{2}\Delta v$ to the positions. Each GP has a squared-exponential ARD kernel with hyperparameters from the marginal likelihood [7], trained on a random subset of 150 of the 1500 samples (a subset-of-data approximation). The constraints are tightened by $\beta = 2$ standard deviations, in the spirit of cautious MPC [4]: the GP variance is evaluated along the shifted previous solution and summed over the horizon, ignoring the Jacobian of the dynamics. *This margin is a heuristic, not a certified bound.*

DeePC. Following [3], the predictor is a Hankel matrix of measured input–output data. With $w_k = (u_k, y_{k+1})$ and $L = T_{\mathrm{ini}} + N$ ($T_{\mathrm{ini}} = 3$), columns from all 15 episodes form a mosaic Hankel matrix [9] (800 columns sampled at random). The QP

$$\min_{g,\sigma}\; J(u,y) + \lambda_g\|g\|^2 + \lambda_\sigma\|\sigma\|^2 \quad \text{s.t.} \quad \begin{bmatrix} U_p \\ Y_p \\ U_f \\ Y_f \end{bmatrix} g = \begin{bmatrix} u_{\mathrm{ini}} \\ y_{\mathrm{ini}} + \sigma \\ u \\ y \end{bmatrix} \tag{2}$$

is solved with OSQP [8]. For a noise-free LTI system with persistently exciting data, the fundamental lemma [11] makes the Hankel predictor exact; a unit test checks this on the mass-spring-damper (error below 10^{-5}). For the nonlinear plants and for noisy data the quadratic regularisation is a heuristic; guarantees for that setting need the extra assumptions and terminal ingredients of [2], which are not implemented here. The weight λ_g was picked from $\{0.01, 0.1, 1, 10, 100\}$ on a separate tuning seed by a fixed rule (Fig. 3).

Koopman MPC. EDMD [12] with inputs, as in [5]: $\psi(x) = (x, 1, \varphi_1(x), \ldots, \varphi_{40}(x))$ with Gaussian radial basis functions on the standardised features, and $\psi(x_{k+1}) \approx A\psi(x_k) + Bu_k$ fitted by ridge least squares. Since x is part of ψ, $x = C\psi$ with $C = [I\ 0]$, and the MPC is a condensed QP solved with OSQP. The case without basis functions is the affine DMDc model [6], used as an ablation.

3 Results

Prediction. Figure 1 and Table 2 show the open-loop prediction error on held-out episodes. All learned models improve on the prior model by one to two orders of magnitude. On the pendulum, which is open-loop unstable, every model drifts after 20 steps; the Hankel predictor drifts least. The radial basis functions improve on the affine DMDc model only on the pendulum; on the mass-spring-damper DMDc is slightly better and on the quadrotor the two are equal to three digits. The GP 2σ interval covered 94–96 % of held-out one-step residuals.

Closed loop. Table 3 gives the metrics over 10 seeds: tracking cost (mean of $\|x - r\|_Q^2$), steady-state error (position error before the reference preview reaches the next set-point), the largest violation (mean over seeds), the number of steps with a violation and the mean solve time.

The main observations, all read from Table 3:

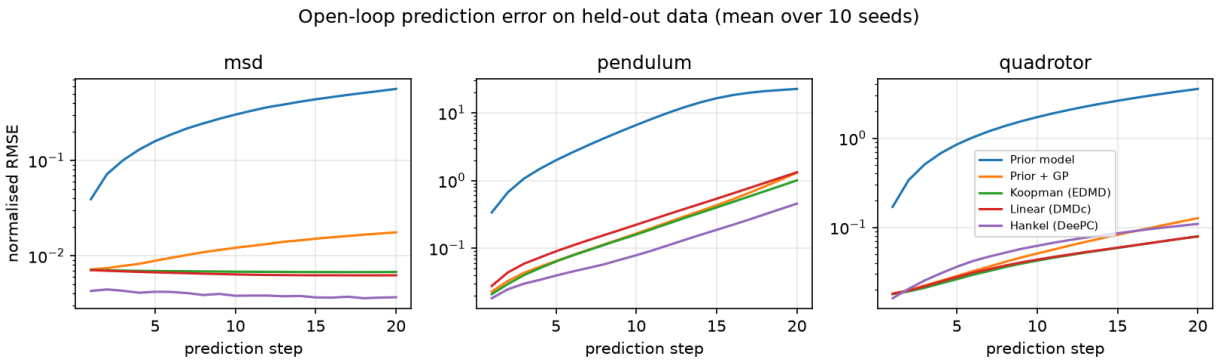


Figure 1: Open-loop prediction error versus horizon (held-out data, mean over 10 seeds).

Table 2: Normalised RMSE at prediction step 20 and GP coverage.

Model	msd	pendulum	quadrotor
Prior model	0.563	22.480	3.562
Prior + GP	0.018	1.306	0.127
Koopman (EDMD)	0.007	1.003	0.079
Linear (DMDc)	0.006	1.320	0.079
Hankel (DeePC)	0.004	0.455	0.109
GP 2σ coverage	0.959	0.951	0.940

- The prior model leaves a steady-state offset (0.042, 0.011 and 0.104 on the three plants). All three learned controllers reduce it: GP-MPC and Koopman MPC to 0.001–0.003, close to the oracle, and DeePC less on the quadrotor (0.041).
- GP-MPC has the cost closest to the oracle on the quadrotor and the fewest violation steps on every plant (on the pendulum tied at zero with nominal MPC and DeePC); it is the only controller that tightens the constraints. It is also the slowest controller on the quadrotor (49.5 ms mean, close to the 50 ms sampling time).
- Koopman MPC is the fastest (below 1 ms) and has the lowest cost on the mass-spring-damper and the pendulum, but the largest violations on the pendulum and the quadrotor (Fig. 2: a sharp roll transient at the start).
- DeePC works with no model at all, but on the quadrotor it has the highest cost and the largest spread across seeds, and on the mass-spring-damper its worst-case solve time (234 ms) exceeds the sampling time.
- The oracle violates the soft constraints slightly on every plant, because of measurement noise and because it has no margin.

4 Limitations

Everything is simulated, with one mismatch scenario per plant and a fixed data budget. The GP uses a subset of the data and its tightening ignores how uncertainty propagates through the dynamics. DeePC and Koopman MPC carry no closed-loop guarantee here. The tracking cost is dominated by the transients after set-point changes, which all controllers share, so differences in cost are small; the steady-state error and the violations separate the methods more clearly.

Table 3: Closed-loop results, mean over 10 seeds (cost: mean ± std).

Controller	Cost	SS error	Max viol.	Steps viol.	Time [ms]
msd					
Nominal MPC	1.021 ± 0.001	0.0420	0.042	11.7	4.7
Oracle MPC	1.017 ± 0.001	0.0006	0.011	20.6	4.8
GP-MPC	1.018 ± 0.002	0.0024	0.006	5.3	9.9
DeePC	1.046 ± 0.006	0.0056	0.006	18.8	15.3
Koopman MPC	1.015 ± 0.002	0.0012	0.012	24.6	0.3
pendulum					
Nominal MPC	0.238 ± 0.001	0.0112	0.000	0.0	5.7
Oracle MPC	0.217 ± 0.000	0.0008	0.007	1.4	6.7
GP-MPC	0.220 ± 0.001	0.0011	0.000	0.0	10.5
DeePC	0.231 ± 0.001	0.0035	0.000	0.0	3.7
Koopman MPC	0.216 ± 0.001	0.0011	0.071	2.7	0.2
quadrotor					
Nominal MPC	1.292 ± 0.010	0.1043	0.021	10.8	9.9
Oracle MPC	1.108 ± 0.003	0.0020	0.005	6.4	10.0
GP-MPC	1.117 ± 0.004	0.0029	0.001	0.5	49.5
DeePC	1.321 ± 0.073	0.0414	0.013	2.5	13.7
Koopman MPC	1.194 ± 0.028	0.0031	0.060	7.7	0.7

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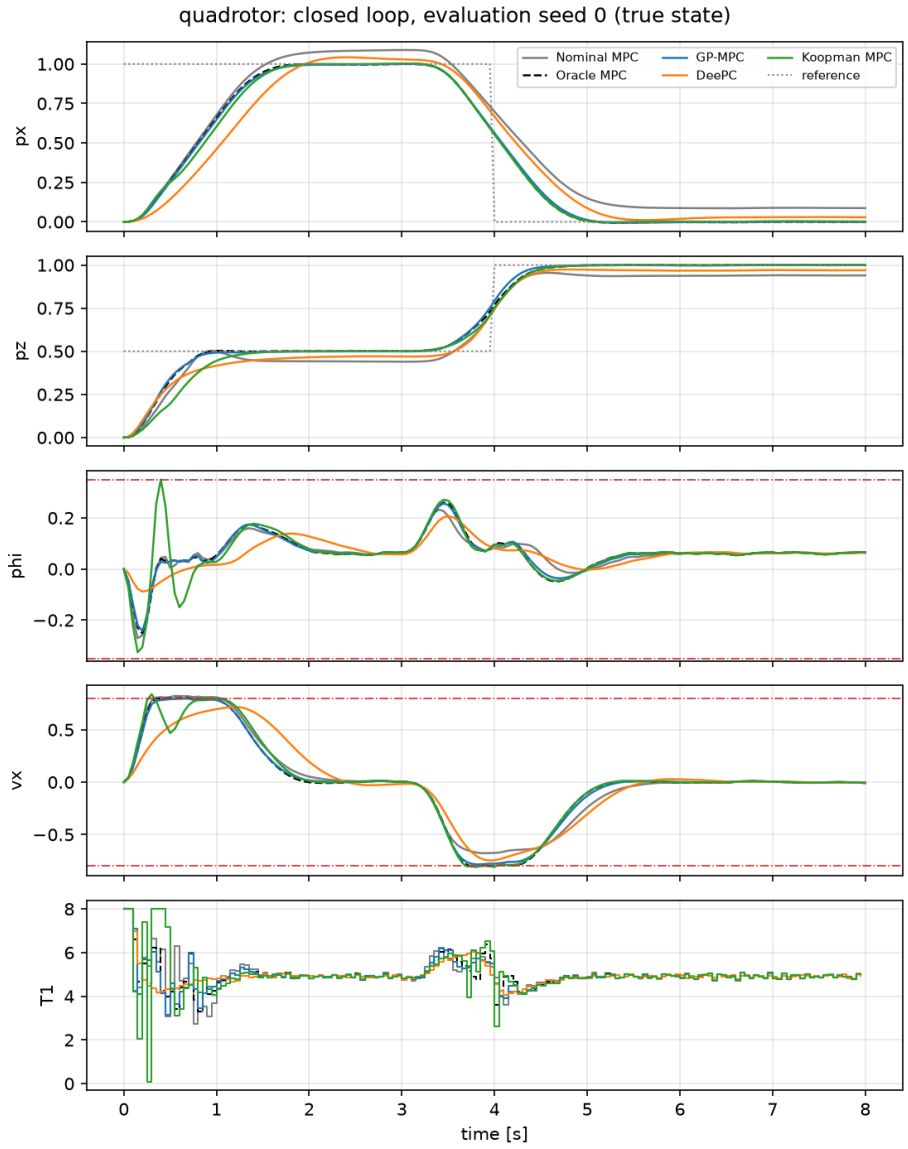


Figure 2: Quadrotor closed loop, evaluation seed 0. Dash-dot lines are the state constraints.

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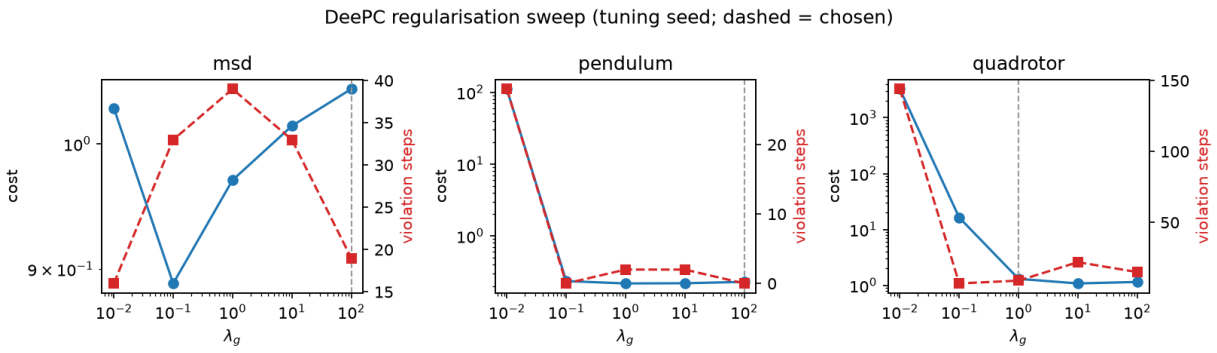


Figure 3: DeePC regularisation sweep on the tuning seed (dashed: chosen weight).